MUHAMMAD ZAIN-UL-ABIDEEN

AI Engineer

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Summary

AI Engineer who ships measured ML and LLM systems end to end: speech search, computer vision, RAG, and agents. Every project is live, benchmarked against baselines on held-out data, and publishes its failure cases alongside its wins.

Education

National University of Sciences and Technology (NUST) Sep 2022 – Jun 2026
Bachelor of Electrical Engineering Islamabad, Pakistan

Technical Skills

LLM & Retrieval: RAG, Agentic Workflows, Hybrid Search (BM25, RRF), Embeddings, Rerankers, Evals (Recall@k, MRR, LLM-as-Judge), Gemini, Groq, Langfuse
ML & Vision: PyTorch, Scikit-learn, Image Classification, Segmentation, CLIP, ONNX Runtime, Selective Prediction, Whisper, Speech
Backend & Data: Python, FastAPI, PostgreSQL, pgvector, REST APIs, SQLAlchemy, Playwright, Next.js
MLOps & Cloud: Docker, GitHub Actions, CI/CD with eval gates, pytest, Git, Render, Vercel, Azure, GCP

Projects

Earshot – Podcast Q&A with Audio Citations | Python, Whisper, pgvector earshot-ten-pi.vercel.app

- Built a podcast Q&A system that transcribes episodes with Whisper large-v3-turbo (2.65% WER, 0.55 s worst word-timestamp error) and links every claim to the second it was said in the original audio.
- Benchmarked keyword, vector, hybrid, and reranked search over 875 passages: a reranker lifted recall@5 from 0.76 to 0.93 (MRR 0.60 to 0.85) at 1.35 s vs. 69 ms latency; shipped hybrid on a 0.1-CPU tier and documented the trade-off.
- Added a verbatim quote checker with a one-shot repair pass that raised answered questions from 77.8% to 93.3% and correctly timed citations from 62.2% to 80.0%.

Dehqan – Crop Doctor for Pakistani Farmers | Python, PyTorch, CLIP, ONNX dehqan.vercel.app

- Trained a 27-condition, 13-crop leaf-disease classifier and exposed its lab-to-field gap (95.5% on lab photos, 11.5% in real fields); a CLIP embedding with a trained head raised held-out field accuracy to 61.6% as deployed in int8 ONNX (63.8% in PyTorch) vs. a 7.4% baseline.
- Shipped selective prediction that defers to extension offices below 0.9 confidence (answers 56% of field photos at 81% accuracy), a zero-shot CLIP leaf gate (99.8% of leaves accepted vs. 23.6% using classifier confidence), and a segmentation severity grader (81.5% vs. 42% baseline).
- Added Urdu/English voice chat (in-browser speech recognition and read-aloud, zero server cost) over a router for English, Roman Urdu, and Urdu questions (90% on a frozen test set) that refused 7/7 pesticide-dose requests.

Rehnuma – AI Copilot for Electricity Bills | Python, FastAPI, Gemini, Groq rehnuma-kappa.vercel.app

- Built an Urdu/English bill auditor where a deterministic engine re-checks every figure a Gemini vision model reads, triggering targeted re-reads (96.2% field accuracy on real photos; 0 invented numbers reached users).
- Developed lexical hybrid RAG (BM25 + character n-grams, RRF) over 566 OCR'd regulatory clauses with a citation critic; query rewriting lifted Urdu top-5 retrieval from 0% to 83% (91% held-out English); 319 tests, 20 eval metrics gated in CI.

Parity – AI Web Accessibility Auditor | Python, FastAPI, Playwright, Gemini parity-iota-puce.vercel.app

- Built a WCAG 2.2 auditor combining axe-core, a pixel-based contrast meter, a keyboard-navigation agent, and a Gemini vision agent; raised labelled-benchmark recall from 54% (rules alone) to 24/24 with zero false alarms.
- Lifted vision-agent precision from 80% to 100% with consistency checks; shipped FastAPI with a job queue, rate limits, and SSRF protection, backed by 190+ tests.

Experience

Machine Learning Engineer Intern Jun 2025 – Aug 2025
RISETech Islamabad, Pakistan

- Led feature extraction for a multi-sensor (vibration, magnetic, acoustic) vehicle classifier; trained ensemble models (Random Forest, SVM, KNN, LDA, Extra Trees) that raised accuracy from 71% to 99%.
- Authored an IEEE-format research paper on the feature extraction method.

AI Engineer Intern Jun 2024 – Aug 2024
Software Productivity Strategists (SPS), NSTP Islamabad, Pakistan

- Developed Python backend features for two flagship products (BMS, CSM) and deployed AI solutions on Microsoft Azure.

Certifications

- Google AI Professional Certificate • Advanced ML on Google Cloud Specialization • IBM Machine Learning Professional Certificate • AWS Cloud Practitioner Essentials